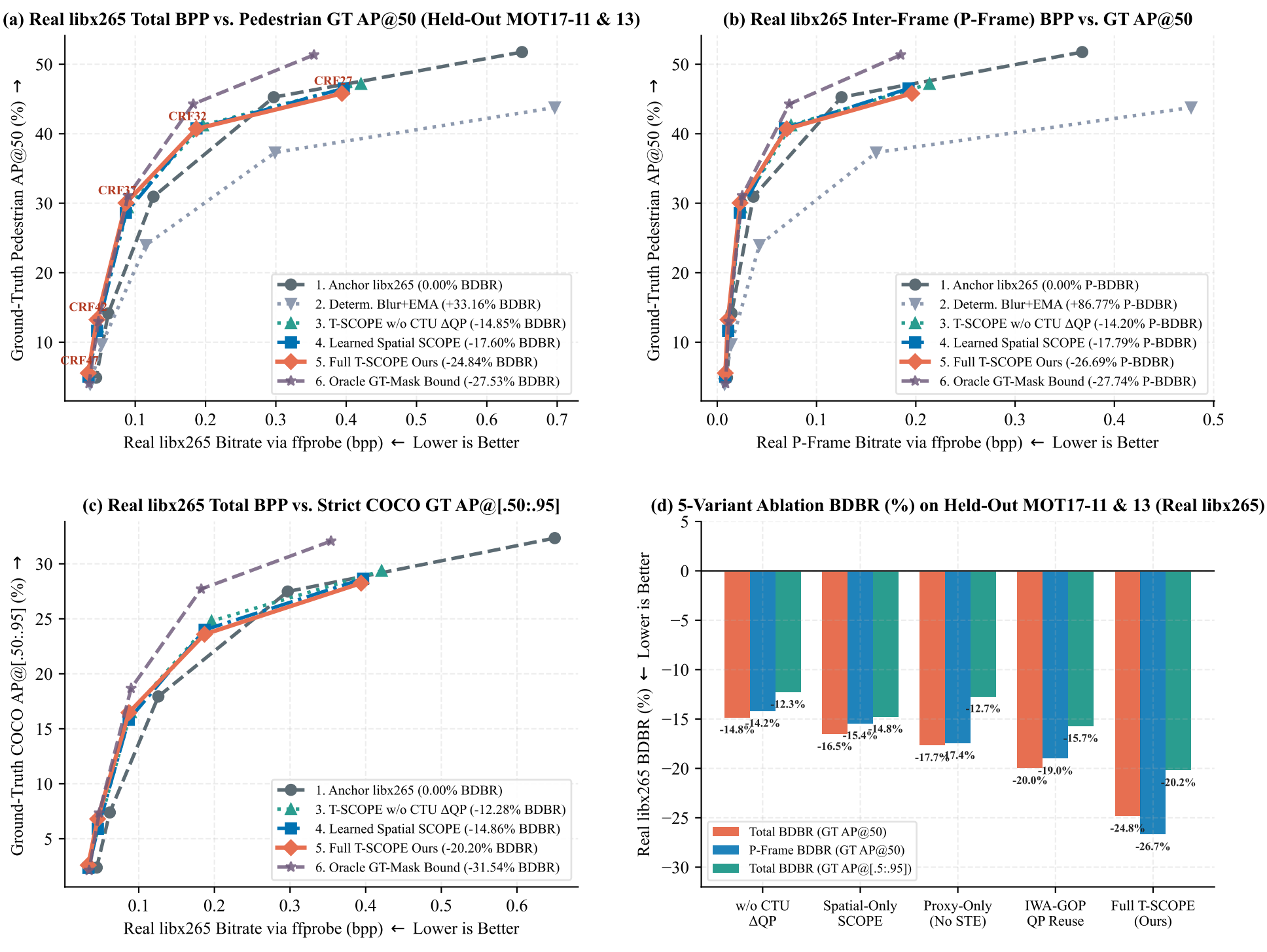

(a) Real libx265 Total BPP vs. Pedestrian GT AP@50 (Held-Out MOT17-11 & 13)
Ground-Truth Pedestrian AP@50 (%) ↑
Real libx265 Bitrate via ffprobe (bpp) ← Lower is Better
CRF27
CRF32
CRF37
CRF42
CRF47
1. Anchor libx265 (0.00% BDBR)
2. Determ. Blur+EMA (+33.16% BDBR)
3. T-SCOPE w/o CTU ΔQP (-14.85% BDBR)
4. Learned Spatial SCOPE (-17.60% BDBR)
5. Full T-SCOPE Ours (-24.84% BDBR)
6. Oracle GT-Mask Bound (-27.53% BDBR)
(b) Real libx265 Inter-Frame (P-Frame) BPP vs. GT AP@50
Ground-Truth Pedestrian AP@50 (%) ↑
Real P-Frame Bitrate via ffprobe (bpp) ← Lower is Better
1. Anchor libx265 (0.00% P-BDBR)
2. Determ. Blur+EMA (+86.77% P-BDBR)
3. T-SCOPE w/o CTU ΔQP (-14.20% P-BDBR)
4. Learned Spatial SCOPE (-17.79% P-BDBR)
5. Full T-SCOPE Ours (-26.69% P-BDBR)
6. Oracle GT-Mask Bound (-27.74% P-BDBR)
(c) Real libx265 Total BPP vs. Strict COCO GT AP@[.50:.95]
Ground-Truth COCO AP@[.50:.95] (%) ↑
Real libx265 Bitrate via ffprobe (bpp) ← Lower is Better
1. Anchor libx265 (0.00% BDBR)
3. T-SCOPE w/o CTU ΔQP (-12.28% BDBR)
4. Learned Spatial SCOPE (-14.86% BDBR)
5. Full T-SCOPE Ours (-20.20% BDBR)
6. Oracle GT-Mask Bound (-31.54% BDBR)
(d) 5-Variant Ablation BDBR (%) on Held-Out MOT17-11 & 13 (Real libx265)
Real libx265 BDBR (%) ← Lower is Better
-14.8%
-14.2%
-12.3%
-16.5%
-15.4%
-14.8%
-17.7%
-17.4%
-12.7%
-20.0%
-19.0%
-15.7%
-24.8%
-26.7%
-20.2%
Total BDBR (GT AP@50)
P-Frame BDBR (GT AP@50)
Total BDBR (GT AP@[.5:.95])
w/o CTU ΔQP
Spatial-Only SCOPE
Proxy-Only (No STE)
IWA-GOP QP Reuse
Full T-SCOPE (Ours)